Simulating Kicked Floquet Ising System

Simulation Details

- **Floquet Ising Model:-**

$$H(t) = J\sum_{i=1}^{L-1} Z_i Z_{i+1} + h_z \sum_{i=1}^{L} Z_i + \frac{\theta}{2}\sum_{n=-\infty}^{\infty} \delta(t - nT) \sum_{i=1}^{L} X_i$$

where L is the number of spins, J_z is the nearest-neighbor Ising interaction strength, h_x is the transverse field, and h_z is the longitudinal field.

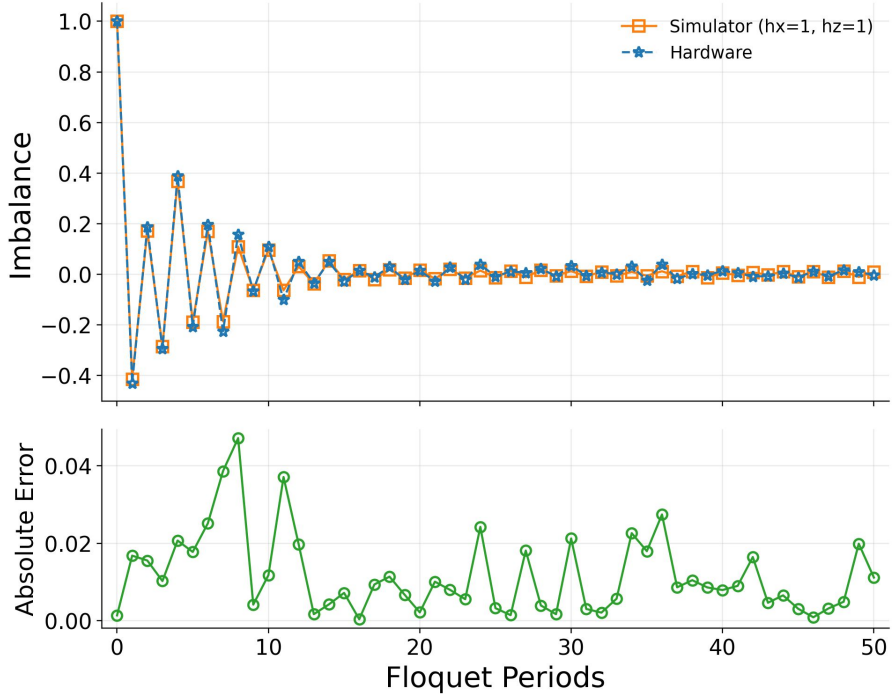
- **Floquet evolution operator:-**

$$U_F = \exp\left(-i\frac{\theta}{2}\sum_{i=1}^{L} X_i\right) \exp\left[-iT\left(J\sum_{i=1}^{L-1} Z_i Z_{i+1} + h_z \sum_{i=1}^{L} Z_i\right)\right]$$

- **Imbalance:-**

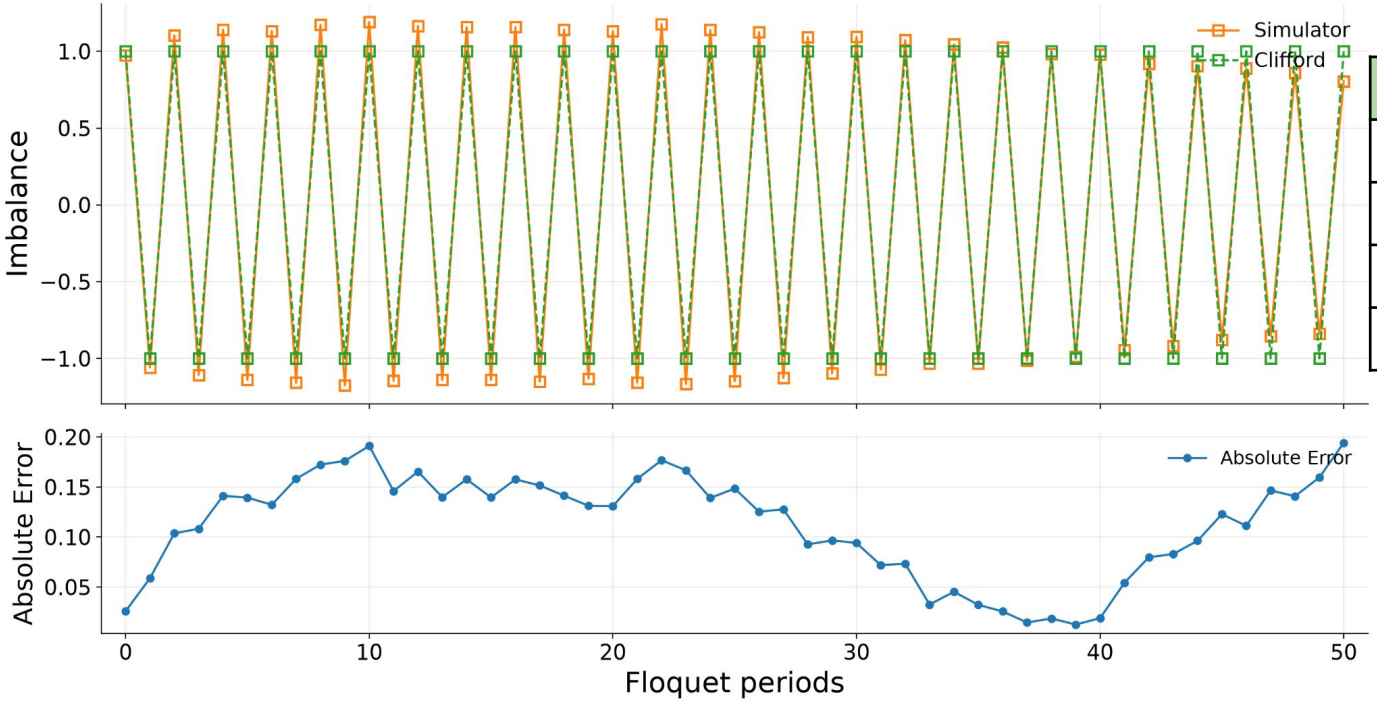
$$\hat{\mathcal{I}} = \frac{1}{L}\sum_{i=0}^{L-1} (-1)^i Z_i$$

Results(N=20)



Parameters	Values
L(num_qubits)	20
Jz	1
hx	1
hz	1

Results(N=100)



Parameters	Values
L(num_qubits)	100
Jz	π/2
hx	π/2
hz	π/2

Things to do and future scope

- Apply different error mitigation strategies like ODR, PNA, etc.
- Compare outcomes of non-clifford circuits with best classical methods like Pauli propagation and tensor network methods
- We can also simulate in 2D in square lattice with latest ibm nighthawk processor

Thank You